`citemsp`: Citation Locators for LaTeX

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Abstract

We present `citemsp`, a LaTeX package that adds per-key locators to numeric citation labels, rendered as compact superscript/subscript pairs[1]. Working on top of either `biblatex` or `natbib`, the package provides a slash-delimited syntax with ten built-in prefix types — covering sections, equations, theorems, figures, and more — and a user-extensible registry for domain-specific locators.

1 Syntax and Usage

The `\citemsp` command accepts a comma-separated list of citation keys with optional locators separated by forward slashes. By default, the first locator slot renders as § (section) and the second as ¶ (paragraph). A single-letter prefix changes the locator type. With no locators, `\citemsp` behaves identically to `\cite`, including range compression when using `numeric-comp` or `sort&compress`.

`\citemsp{key}`	$\longrightarrow$	$[1]$
`\citemsp{key/section}`	$\longrightarrow$	$[1^{\S 6.2}]$
`\citemsp{key/section/paragraph}`	$\longrightarrow$	$[1^{\S 6.2}_{\P 3}]$
`\citemsp{key//paragraph}`	$\longrightarrow$	$[1_{\P 3}]$
`\citemsp{key/section/e4.3}`	$\longrightarrow$	$[1^{\S 6.2}_{\text{eq.}4.3}]$

Currently there exist ten built-in prefixes, which are listed in Table 1. Each prefix can appear in either the superscript or subscript slot. Users can register additional prefixes with `\citemspprefix{letter}{label}`.

2 Examples

[1]The package is available at https://github.com/ApostolosTsabodimos/citemsp-Latex-package and on CTAN.

Prefix	Locator	Render	In super slot	In sub slot	Both slots
`A`	Appendix	App.	$[1^{\text{App.}\,2}]$	$[1_{\text{App.}\,2}]$	$[1^{\S 3.2}_{\text{App.}\,2}]$
`C`	Corollary	Cor.	$[1^{\text{Cor.}\,3}]$	$[1_{\text{Cor.}\,3}]$	$[1^{\S 4.1}_{\text{Cor.}\,3}]$
`d`	Definition	$\triangleq$	$[1^{\triangleq 5}]$	$[1_{\triangleq 5}]$	$[1^{\S 3.2}_{\triangleq 5}]$
`e`	Equation	eq.	$[1^{\text{eq.}\,4.3}]$	$[1_{\text{eq.}\,4.3}]$	$[1^{\S 6.2}_{\text{eq.}\,4.3}]$
`f`	Figure	fig.	$[1^{\text{fig.}\,3}]$	$[1_{\text{fig.}\,3}]$	$[1^{\S 6.1}_{\text{fig.}\,3}]$
`L`	Lemma	Lem.	$[1^{\text{Lem.}\,1}]$	$[1_{\text{Lem.}\,1}]$	$[1^{\S 3.2}_{\text{Lem.}\,1}]$
`n`	Footnote	†	$[2^{\dagger 7}]$	$[2_{\dagger 7}]$	$[2^{\S 1}_{\dagger 7}]$
`p`	Page	pp.	$[1^{\text{pp.}\,42}]$	$[1_{\text{pp.}\,42}]$	$[1^{\S 3.2}_{\text{pp.}\,42}]$
`t`	Table	$\boxplus$	$[3^{\boxplus 2}]$	$[3_{\boxplus 2}]$	$[3^{\S 8.4}_{\boxplus 2}]$
`T`	Theorem	Th.	$[1^{\text{Th.}\,4.2}]$	$[1_{\text{Th.}\,4.2}]$	$[1^{\S 6.1}_{\text{Th.}\,4.2}]$

Table 1: Built-in prefix registry.

2.1 Basic Usage

```
The Schwarzschild metric~\citemsp{wald1984/6.1/2} describes
spherically symmetric vacuum solutions.
```

Output: The Schwarzschild metric $[1^{\S 6.1}_{\P 2}]$ describes spherically symmetric vacuum solutions.

Multiple sources with independent locators:

```
Black hole thermodynamics~\citemsp{hawking1974/1/2,
wald1984/12.5/1} establishes a deep connection between
gravity and quantum mechanics.
```

Output: Black hole thermodynamics $[2^{\S 1}_{\P 2}, 1^{\S 12.5}_{\P 1}]$ establishes a deep connection between gravity and quantum mechanics.

Locator entries can be freely mixed:

```
General covariance~\citemsp{einstein1915, wald1984/4,
hawking1974/1/2} forms the foundation of general relativity.
```

Output: General covariance $[4, 1^{\S 4}, 2^{\S 1}_{\P 2}]$ forms the foundation of general relativity.

2.2 Similarity to \cite

With no locators, `\citemsp` passes keys through the backend's native `\cite`, producing identical output:

`\cite{wald1984}` $\longrightarrow$ [1] `\citemsp{wald1984}` $\longrightarrow$ [1]

All plain (5 consecutive): [1–5]

All plain (non-consecutive): [1, 4–8]

Range, then locator: $[1,\ 4,\ 5, 2^{\S 3.2}]$

Locator, then range: $[4^{\S 3.2}, 1\text{–}3,\ 5]$

Range, locator, range: $[1,\ 4,\ 5, 2^{\text{eq.}\,2.1}, 3,\ 6,\ 7]$

Adjacent locators only: $[1^{\S 3.1}, 2^{\text{eq.}\,2.1}, 3^{\text{pp.}\,42}]$

2.3 Mixed Types and Custom Prefixes

All built-in locator types can be freely mixed in a single call:

```
See~\citemsp{wald1984/T6.2, einstein1915/3.2/d5,
mtw1973/p520, hawking1974/1/e4.3} for further details.
```

Output: See [$1^{\text{Th.}6.2}$, $4^{\S 3.2}_{\triangleq 5}$, $3^{\text{pp.}520}$, $2^{\S 1}_{\text{eq.}4.3}$] for further details.

Users can register domain-specific prefixes with `\citemspprefix`. For example, a cosmology paper might define prefixes for window functions and sky patches:

```
\citemspprefix{w}{w.}
\citemspprefix{S}{\ensuremath{\Omega}}
The top-hat window~\citemsp{wald1984/4.2/w3} smooths
the power spectrum over sky patch~\citemsp{mtw1973/S12}.
```

Output: The top-hat window [$1^{\S 4.2}_{\text{w.}3}$] smooths the power spectrum over sky patch [$3^{\Omega 12}$].

2.4 Paragraph example

The following paragraph illustrates how `\citemsp` reads in running text, mixing plain citations, section locators, and prefix types:

```
The Schwarzschild solution~\citemsp{schwarzschild1916/1/3}
was found shortly after the field
equations~\citemsp{einstein1915} were published.  A modern
treatment can be found in standard
textbooks~\citemsp{wald1984/6.1/2, mtw1973/p520}, where the
derivation proceeds via the Birkhoff
theorem~\citemsp{wald1984/T6.2}.  The singularity
theorems~\citemsp{penrose1965, hawking1974/1/e4.3} later
showed that gravitational collapse is generic.  For further
context see~\citemsp{kerr1963, chandrasekhar1931}.
```

Output: The Schwarzschild solution [$5^{\S 1}_{\P 3}$] was found shortly after the field equations [4] were published. A modern treatment can be found in standard textbooks [$1^{\S 6.1}_{\P 2}$, $3^{\text{pp.}520}$], where the derivation proceeds via the Birkhoff theorem [$1^{\text{Th.}6.2}$]. The singularity theorems [6, $2^{\S 1}_{\text{eq.}4.3}$] later showed that gravitational collapse is generic. For further context see [7, 8].

3 Loading and Configuration

With biblatex

```
\usepackage[style=numeric-comp, backend=biber]{biblatex}
\addbibresource{refs.bib}
\usepackage{citemsp}
```

Use `numeric-comp` (or `numeric`) as the style. Range compression of consecutive plain entries requires `numeric-comp`.

Configuration

The locator glyph size is controlled by `\citemspscale` (default 0.35):

Default (0.35): [$1^{\S 6.1}_{\text{eq.}4.3}$] Smaller (0.25): [$1^{\S 6.1}_{\text{eq.}4.3}$] Larger (0.50): [$1^{\S 6.1}_{\text{eq.}4.3}$]

To adjust the locator glyph size, redefine `\citemspscale` after loading:

```
\usepackage{citemsp}
\renewcommand{\citemspscale}{0.4}  % default is 0.35
```

4 Relation to Standard Locators

The standard `biblatex` provides locators via its postnote mechanism, the \citemsp package suggests a more compact mechanism:

```
% Standard biblatex:
\autocites[§6.1, ¶2]{wald1984}[pp.\,520]{mtw1973}[eq.\,4.3]{hawking1974}

% citemsp equivalent:
\citemsp{wald1984/6.1/2, mtw1973/p520, hawking1974//e4.3}
```

Three advantages distinguish `\citemsp`: (1) a compact, uniform syntax that stays readable as entries accumulate; (2) typed locator prefixes that visually distinguish sections, equations, pages, theorems, and other reference types; and (3) superscript/subscript rendering that keeps locator information close to each citation number rather than appended as flat text.

5 Conventions

- **Section numbering.** Use the source's native section numbering format (e.g., `6.1` for "Section 6.1").
- **Paragraph counting.** Paragraphs are counted manually from the start of the section. This is inherently edition-dependent and best suited for stable references such as textbooks.
- **Custom prefixes.** Register new locator types with `\citemspprefix{letter}{label}`. The letter must be a single character not already in use. Custom and built-in prefixes can be freely mixed in the same `\citemsp` call.

References

[1] Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.

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[3] Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler. *Gravitation*. W. H. Freeman, 1973.

[4] Albert Einstein. "Die Feldgleichungen der Gravitation". In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1915), pp. 844–847.

[5] Karl Schwarzschild. “Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie”. In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1916), pp. 189–196.

[6] Roger Penrose. “Gravitational Collapse and Space-Time Singularities”. In: *Physical Review Letters* 14 (1965), pp. 57–59.

[7] Roy P. Kerr. “Gravitational Field of a Spinning Mass as an Example of Algebraically Special Metrics”. In: *Physical Review Letters* 11 (1963), pp. 237–238.

[8] Subrahmanyan Chandrasekhar. “The Maximum Mass of Ideal White Dwarfs”. In: *The Astrophysical Journal* 74 (1931), pp. 81–82.